

Sungyeon Kim

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SUMMARY

I am an Applied Scientist II at AWS Agentic AI, where I work on AgentCore Evaluation and Strands Agents. I received my Ph.D. in Computer Science and Engineering from POSTECH, advised by Prof. Suha Kwak. At AWS, my work spans agent evaluation, reliability signals, evaluation methodology, and multimodal evaluators. Previously at Amazon Core Search, I contributed to Amazon Lens Live through multimodal retrieval, result verification, and MLLM-based re-ranking. My research focuses on reliable agentic AI systems, agent evaluation, agentic retrieval, multimodal retrieval, MLLM-based retrieval/ranking, long-horizon LLM agents, agent memory and context management, and representation learning, with side interests in VLA robot learning and RL (PPO/GRPO).

WORK EXPERIENCE

Amazon Web Services – Agentic AI

Oct 2025 – Present

Applied Scientist II

Santa Clara, CA

- Core contributor for Multimodal Evaluators in Strands Evals and AgentCore Evaluation; co-led first-party evaluation model training and evaluation for agent evaluation.

Amazon.com – Core Search

Aug 2025 – Oct 2025

Applied Scientist II

Palo Alto, CA

- Contributed multimodal retrieval, verification, and MLLM-based re-ranking for Amazon Lens Live.

Amazon.com – Core Search

Jun 2024 – Sep 2024

Applied Scientist Intern

Palo Alto, CA

- Developed GENIUS, a generative retrieval framework for universal multimodal search; published at CVPR 2025.

SKILLS

Focus Areas: Agentic AI, agent evaluation, agentic retrieval, multimodal retrieval, MLLM-based retrieval/ranking, long-horizon LLM agents, agent memory and context management; side: VLA robot learning, RL, PPO/GRPO.

Research: 15 publications at CVPR, ICCV, and ECCV on multimodal retrieval, deep metric learning, domain generalization, and representation learning; 1.2k+ citations.

Skills: Python, PyTorch (Advanced), JAX (Basic), AWS Bedrock, vLLM, LiteLLM, Hydra, Slurm, Git; Languages: Korean (Native), English (Professional Working Proficiency).

SELECTED PUBLICATIONS

- Sungyeon Kim** et al., “GENIUS: A Generative Framework for Universal Multimodal Search,” *CVPR 2025*.
- Boseung Jeong et al., “Learning Audio-guided Video Representation with Gated Attention for Video-Text Retrieval,” *CVPR 2025, Oral Presentation*.
- Sungyeon Kim** et al., “Efficient and Versatile Robust Fine-Tuning of Zero-shot Models,” *ECCV 2024*.
- Sungyeon Kim** et al., “HIER: Metric Learning Beyond Class Labels via Hierarchical Regularization,” *CVPR 2023*.
- Sungyeon Kim** et al., “Self-Taught Metric Learning without Labels,” *CVPR 2022*.
- Sungyeon Kim** et al., “Proxy Anchor Loss for Deep Metric Learning,” *CVPR 2020, 580+ citations*.
- Sungyeon Kim** et al., “Deep Metric Learning Beyond Binary Supervision,” *CVPR 2019, Oral Presentation*.

EDUCATION

POSTECH (Pohang University of Science and Technology)

Pohang, Republic of Korea

Ph.D. in Computer Science and Engineering

Sep 2018 – Feb 2025

- Advisor: Prof. Suha Kwak.

Dissertation: *Towards Retrieval at Scale via Compact Embeddings and Generative Modeling*.

DGIST (Daegu Gyeongbuk Institute of Science and Technology)

Daegu, Republic of Korea

B.S. in School of Undergraduate Studies

Mar 2014 – Feb 2018

HONORS AND SERVICE

Honors: Google PhD Fellowship, 2023; Qualcomm Innovation Fellowship Korea, 2020, 2022, 2024.

Service: Reviewer for CVPR, ICCV, ECCV, ICLR, ICML, NeurIPS, AAAI, ACCV, TPAMI, IJCV.